Endurance's sky. Truly awe inspiring, but undesirable for so early in the movie! So Chris and Paul chose to make Gargantua look much smaller at the parking orbit: about two and a half degrees, which is five times the size of the Moon as seen from Earth—still impressive but not overwhelmingly so.

The Shell of Fire



Gravity is so strong near Gargantua, and space and time are so warped, that light (photons) can be trapped in orbits outside the horizon, traveling around and around the hole many times before escaping. These trapped orbits are *unstable* in the sense that the photons always escape from them, eventually. (By contrast, photons caught inside the horizon can never get out.)

I like to call this temporarily trapped light the "shell of fire." This fire shell plays an important role in the computer simulations (Chapter 8) that underlie Gargantua's visual appearance in *Interstellar*.

For a *non*spinning black hole, the shell of fire is a sphere, one with circumference 1.5 times larger than the horizon's circumference. The trapped light travels around and around this sphere on great circles (like the lines of constant longitude on the Earth); and some of it leaks into the black hole, while the rest leaks outward, away from the hole.

When a black hole is spun up, its shell of fire expands outward and inward, so it occupies a finite volume rather than just the surface of a sphere. For Gargantua, with its huge spin, the shell of fire in the equatorial plane extends from the bottom red circle of Figure 6.3 to the upper red circle. The shell of fire has expanded to encompass Miller's planet and the critical orbit, and much, much more! The bottom red circle is a light ray (a photon orbit) that moves around and around